Eco-Evo-Devox: A Modular Artificial Life Platform for Studying the Ecological and Developmental Roots of Neural Evolution

Erwan Plantec[1], Sebastian Risi[1], and Ricard Solé[2,3]

[1]Robotics Evolution and Artificial Life lab, IT University, Copenhagen, Denmark
[2]Complex Systems Lab, ICREA/UPF, Barcelona, Spain
[3]Santa Fe Institute, Santa Fe, New Mexico, USA
{erpl,sebr}@itu.dk

Abstract

Neurons and neural circuits are key evolutionary innovations underlying complex behaviour and cognition. From basic sensorimotor cells to intricate interneuronal networks, their emergence marks a major evolutionary transition in biological complexity. However, the origins of neural circuits and the relative role of development and ecology in their evolution remain poorly understood. Two main theoretical frameworks have been proposed to explain this process. One is based on changes in developmental programs while the second favours environmental factors. However, their interaction may better explain neural evolution. We present *Eco-Evo-Devox* (EEDx), an artificial life platform designed to study these combined factors. EEDx enables expressive, code-free ecological modelling and supports developmental neural networks, leveraging hardware acceleration to simulate large-scale environments and neural ontogeny.

Data/Code available at: `https://github.com/erwanplantec/ecoevodevox`

Introduction

The rise of neural agents marks a turning point in the evolution of complexity in our biosphere (16; 34; 10; 14). Their emergence required several preconditions, including an ecological context, novel developmental programs capable of generating diverse forms, and a new class of cells (neurons) necessary to allow movement in space and deal with uncertainty (19; 32). Paleobiological and molecular evidence shows that around 550 Myr ago, in what is known as the *Cambrian explosion* (CE), a fast diversification of complex organisms occurred (11; 7). This was a revolutionary event, the most important in the history of life in the last billion years. All animal body plans arise by then. How?

Multicellular animal life was already present before the Cambrian. However, this fauna *Ediacaran* lacked the diversity and complexity of the Cambrian and was dominated by sessile organisms (13). Several hypotheses have been advanced regarding the nature of the CE event. Ecologically, changes such as increased oxygen levels, new predator-prey dynamics, and the expansion of ecological niches are proposed to create conditions that favour rapid evolution and diversification (22; 28; 8). Secondly, the evolution of new gene regulatory networks and developmental mechanisms may have enabled more complex body plans (31). Finally, it is suggested that nervous systems and sensory organs, particularly vision, led to complex cognitive capacities (17; 25).

None of these potential contributions to the CE event is independent of the others (12; 30). Changes in the genetic toolkit would have allowed for novel ecological interactions, such as predation. These early predators would have evolved sensors and movement, which required neural integration (and brains), and also triggered an arms race between prey and predators. New developmental innovations unfolded and new ecological niches were created. The tight connection between these three components makes it difficult to weigh their relative impacts and synergies. A powerful way to disentangle them is grounded in an eco-evo-devo artificial life approach, where agents incorporating developmental and cognitive dimensions can coevolve with their synthetic environments.

Simulation architectures have been developed to explore this scenario. EcoEvoJax is a JAX-based simulation platform (5) designed for the study of eco-evolutionary dynamics in large non-episodic multiagent systems (15). The environment in this system is a grid with autonomously growing resources, and agents' behaviours are modelled with directly encoded recurrent neural networks. Recently, Tiwary et al. (35) proposed a platform for studying the evolution of vision, emphasising ecological pressures. In this system, the physical eye structure and a directly encoded neural architecture are evolved in various episodic environments, embedding different pressures, allowing for the exploration of various ecologically driven bifurcations in eye evolution.

In this paper, we present a novel platform, Eco-Evo-Devox (EEDx), a JAX-based non-episodic multi-agent simulation platform for studying the ecological, developmental and evolutionary roots of neural systems. Here, agents have to consume resources in order to maintain their energy levels above a threshold allowing them to reproduce. EEDx incorporates a simple and scalable yet expressive environmental model that allows one to tune and create various ecological contexts capturing key properties of the sensory ecology of early neural evolution where available resources and information landscapes went through important changes (28). While the environment is modelled as a grid, like in Hamon et al. (15), agents in EEDx move in a continuous space, allowing for an embodied connection to the environment that facilitates the emergence of fine locomotor control (see Chaturvedi et al. (9) for another multiagent system with continuous control dynamics).

EEDx allows us to study the effect of embodiment-induced evolutionary pressures by making sensory (i.e. how the nervous system interfaces with the environment) and motor (i.e. how nervous systems induce movement) interfaces easily changeable, from ciliated organisms like embodiments, to Braitenberg vehicles (6), to discrete agents moving on the grid. Where the evolution of nervous systems present an interesting economic conundrum in terms of energy, it appears important to be able to model this trade-off. EEDx allows for that by including energetic costs of neural processing, locomotion, and sensing and varying energy returns from the environment.

Crucially, EEDx differs from previous systems in that agents in this system incorporate a developmental layer that generates their neural phenotype from their inherited genetic material. Complementing existing models like HyperNEAT (33), HyperNCAs (23) or Neural Developmental Programs (NDPs) (24; 27), we propose RAND (*Regulation based Artificial NeuroDevelopment*), a new model for developing artificial neural networks inspired by the gene regulation dynamics found in biological neural networks.

Environment

The environment in EEDx is modelled as a grid $\Omega \subset \mathbb{N}^2$. It is populated with agents whose only objective is survival and reproduction. To do so, agents must maintain their energy level above a certain threshold by consuming food. Agents can perceive food sources and other agents via chemicals (e.g. odors) emitted by the food sources and agents, respectively. The environment can contain multiple types of food and chemicals, each with different properties, allowing for fine-grained design of the environmental dynamics. In the following sections, we describe in detail the resource and chemicals models. For each simulation component, we briefly describe their software implementation and explain how they can be tuned via a simulation configuration file (YAML file) that allows users to create simulations and experiments even with no programming knowledge. Finally, we highlight the potential of this environment model to represent different ecological characteristics, and their transition, which have been hypothesized to be very influential in the early neural evolution process.

Resource Dynamics

Multiple resources can co-exist in EEDx. We note the set of different types of resources, or foods, as $\mathbb{F} \equiv \{F_f\}_{0<f<n^F}$ where each type of food corresponds to a set of parameters that control its energy content, its growth dynamics and its chemical diffusion properties. The presence of different types of food on the grid is given by the food field $\mathcal{F} : \Omega \rightarrow \{0,1\}^{n^F}$

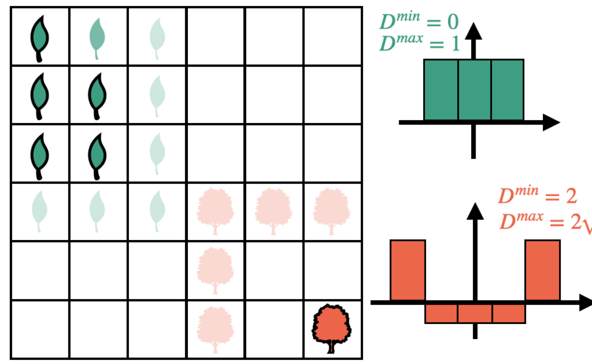


Figure 1: Food growth dynamics in EEDx. The figure illustrates an environment with two food types, each associated with a different growth kernel as shown on the right side. Opaque items with a black border indicate the presence of food in the cell, while the transparent items indicate a non-zero growth probability.

Growth Dynamics The growth dynamics in the EEDx are modelled via convolution operations defining a growth probability field at time $\mathcal{G}^t : \Omega \rightarrow [0,1]^{n^F}$ associating to each cell x the probability that each food type grows there, i.e. $\mathbb{P}(\mathcal{F}^{t+1}(x) \leftarrow 1) = \mathcal{G}^t(x)$. Each of the food growth fields is independent and the growth probability field for food type f, $\mathcal{G}_f$, is given by the two-dimensional convolution of the food field $\mathcal{F}_f$ with the growth convolution kernel G_f defined as :

$$G_f(x) = \begin{cases} \frac{1}{\zeta_f}\rho_f \textbf{ if } D_f^{min} \leq ||x||_2 \leq D_f^{max} \\ -1 \textbf{ else} \end{cases} \quad (1)$$

With ρ_f, the growth rate of food type f. ζ is a normalization constant ensuring $\int G_f = \rho_f$. Food types also have a spontaneous growth parameter $\rho_f^{spontaneous}$ that defines the probability that food of type f is added spontaneously to a cell. This gives us the equation for the growth field:

$$\mathcal{G}_f^t = \text{clip}(\mathcal{F}_f^t * G_f + \rho_f^{spontaneous},\ 0,\ 1) \quad (2)$$

Finally, a Bernoulli trial is simulated in each cell to decide whether food is added to it if it was previously empty. If there was already food in the cell, nothing changes.

Implementation and usage By using convolution operations to model growth dynamics, EEDx takes advantage of the great parallelisation capabilities offered by modern hardware like GPUS or TPUS. Furthermore, we implement convolution using FFT-based convolution operation (29) which is very efficient and allows for larger kernels than the usual 3×3 convolution kernels at no extra cost, allowing greater modelling freedom. The definition of the different types of food and their properties in the configuration file is fairly straightforward, as shown in Listing 1: each type of food is defined by adding an entry in the YAML configuration of the form ft-{x} where x can be any string of characters. Under this type denomination are added the specific parameters of the food type: the growth rate (ρ), minimal and maximal growth distances (D^{min} and D^{max} respectively), the chemical signature (ζ, see next section), the energy content (see Agents section), its spontaneous growth probability ($\rho^{spontaneous}$) and its initial density defining the probability of a cell containing this food type at time $t = 0$.

Listing 1: Food types configuration

```
ft-x:
    growth_rate: 0.01
    dmin: 1
    dmax: 2
    chemical_signature: 0
    energy_concentration: 1.0
    spontaneous_grow_prob: 0.0
    initial_density: 1.0
```

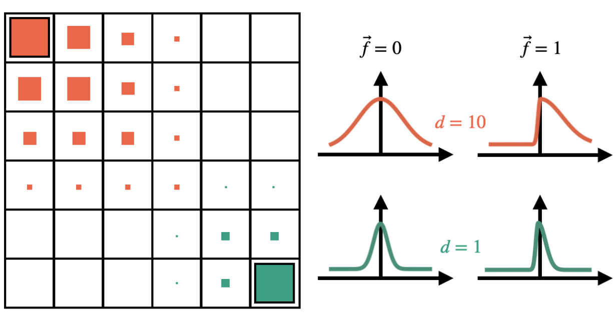


Figure 2: Chemicals diffusion in EEDx. Illustrated is the chemical field $\mathcal{C}$ with two chemical species. Both species have a single source (coloured squares with black contour). The chemicals have different diffusion coefficients, giving different diffusion kernels shown on the right (on a single dimension). On the right are shown kernels in the presence of a flow.

Sensory Landscape

In EEDx, agents perceive the environment by sensing different chemical concentrations where $\mathbb{C} \equiv \{C_c\}_{0<c<n^c}$ is the set of chemical types. Chemicals can be emitted by (i) food sources and (ii) agents. Each type of food is associated with a chemical signature, $\zeta_f \in [0,1]^{n^C}$ for food type f, defining which chemicals it emits and in what proportion. Agents are also associated with a chemical signature: $\zeta_a \in [0,1]^{n^C}$ for agent a. Together, agents and food sources create a chemical source field $\tilde{\mathcal{C}}^t : \Omega \to \mathbb{R}_+^{n^C}$ which is computed as follows:

$$\tilde{\mathcal{C}}^t(x) = \sum_f \mathcal{F}_f^t(x) \cdot \zeta_f + \sum_a [x \in B(a))] \cdot \zeta_a \quad (3)$$

Where $B(a)$ is the set of cells covered by agent a's body (Figure 5). Chemicals then diffuse into the environment via a two-dimensional convolution of the source field with diffusion kernels $\{D_c\}$. In the absence of flow in the environment, the diffusion kernels take the form of two-dimensional radially symmetric Gaussian kernels whose width is defined by chemical type diffusion coefficients d_c:

$$D_c(x) = \exp\left(-\frac{||x||^2}{d_c}\right) \quad (4)$$

In the presence of a nonzero flow in the environment, is modelled with radially asymmetrical diffusion kernels whose width parameter depends on the direction. Let $\vec{f} = \in \mathbb{R}^2$ be the flow in the environment, the diffusion kernel D_c is given by replacing the constant width d_c with a direction dependant width:

$$\mathrm{d}_c(x, \vec{f}) = S(x, \vec{f}) \cdot d_c \cdot ||\vec{f}|| + (1 - S(x, \vec{f})) \cdot d_{min} \quad (5)$$

Where $S(a, b) \in [0, 1]$ is the normalized cosine similarity function ($S(a, b) = \frac{1}{2}\big(\frac{a \cdot b}{||a|| \cdot ||b||} + 1\big)$) and $d_{min} \in \mathbb{R}$ is the minimum diffusion coefficient.

Given diffusion kernels D_c, the chemical field after diffusion $\mathcal{C} : \Omega \to \mathbb{R}_+^{n^C}$ is given by taking the two-dimensional channelwise convolution of the source field with diffusion kernels:

$$\mathcal{C}_c^t = \tilde{\mathcal{C}}_c^t * D_c \quad (6)$$

At larger scales, chemicals are transported by bulk flow creating patchy and intermittent signals (28). While simulating odour plumes dynamics would be expensive, we emulate this patchiness and intermittence by treating the smooth landscape described above, multiplied by a chemical type specific base emission rate e_c, as the probability of an odour signal being present in each cell. We then attribute a value in $\{0, 1\}$ to each cell by performing an independent Bernoulli trial at each time step: $\mathbb{P}(\mathcal{C}_c^t(x) = 1) = (\tilde{\mathcal{C}}_c^t * D_c)(x) \times e_c$.

Implementation and Usage While this model does not reproduce the dynamics of turbulence-induced complex odor plumes (36), it strikes a good balance between expressivity and scalability. Its convolutional formulation allows it to take a large advantage of hardware acceleration. As with food kernels, we use FFT-based convolutions. Listing 2 shows how a chemical type is setup within the configuration file as well as a global flow (here $\vec{v} = (0, 1)$). Sparse diffusion corresponds to probabilistic sampling emulating intermittent odour plumes.

Listing 2: Chemical types configuration

```
flow: [0,1]
ct-x:
  diffusion_rate: 1.0 # gaussian std
  is_sparse: true     # if masked
  emission_rate: 0.01
```

An Ecological Context for Early Neural Evolution

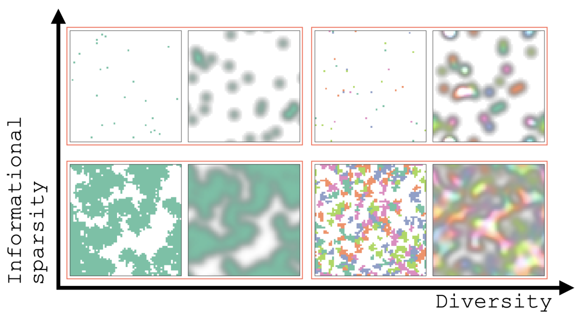


Figure 3: Axis of variation in the sensory ecology of early neural evolution. We show four environments where each pair of images show the resource (left) and chemical (right) landscapes.

The Ediacaran-Cambrian transition is characterized by profound environmental changes associated with the advent of bilateria. These changes notably include the apparition of macroscopic predation, an increase in the size and energy content of organisms as well as the disruption of matgrounds via increased bioturbation, i.e. the perturbation of habitat by caused by mobile organisms (28). These changes led to sharp transition in the structure of *information landscapes*, i.e. the spatio-temporal distribution of available biologically relevant signals (28). In particular, there was likely an increase in the patchiness, or sparsity, of these landscapes caused by seafloor disruption and the concentration of carbon sources in larger animals. Where acquiring and using information about its environment is crucial for survival, this increased complexity of information landscapes might have represented an important pressure for the evolution of more complex sensory, motor and cognitive capabilities, an evolutionary landscape that might have allowed nervous systems to thrive. An other important axis of variation associated with the Cambrian radiation has been the increase in diversity of relevant biological signals also contributing to the overall increase in complexity of information landscapes.

It appears thus interesting to be able to control experimentally the information landscape's structure in order to test early neural evolution theories involving them. EEDx allows to emulate information landscape transformations by tuning the different food and chemical species, and their properties present in the environment. For instance, the increase in the diversity of food types associated with different chemicals can emulate the increase in signal diversity during the CE. Complementarily, environment sparsity can be tuned by varying growth range parameters $\{D^{min}, D^{max}\}$ where larger values will lead to sparse resource fields reminiscent of savannah-like environments (8). Examples of different environments along these axis of variation are shown in Figure 3. Interesting information can also be gained by studying the effect of transitions between different conditions. We facilitate this by providing utilities to simulate transition events. This can be achieved using scenario configuration files (see Listing 3) where multiple environment types, each with their own configuration file can be put together by specifying their start and times.

Listing 3: Scenario configuration file structure

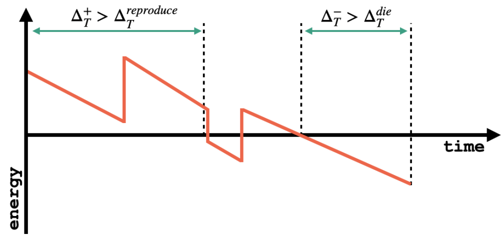
```
env-x:
    file: "my_config_file.yml"
    from: 0      #start time
    to: 10_000 #end time
    on_start = "init"
```

Agents

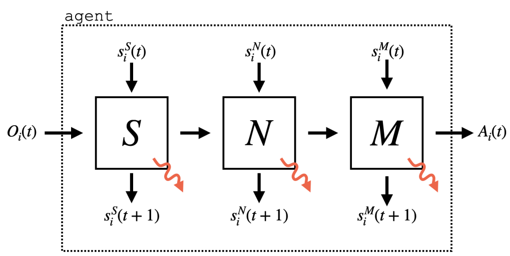
Agents' behaviours are defined by three interacting components: a sensory interface (i) transforming local environmental states into inputs for the agent's neural network (ii), a neural state that is decoded into actions by the motor interface (iii). Importantly, neural networks in EEDx can undergo a developmental phase where a genotype, inherited from the parent agent and mutated, is transcribed into a neural network (the phenotype). In the following sections, we describe these components in detail, presenting their specific instantiations, together with their general formalism. EEDx has been built with modularity in mind, making it easy to plug different models together with no changes to the source code. For instance, if one wants to test a different motor interface not provided by default, custom components can be registered and used very simply by respecting given signatures.

Energy, Reproduction and Death

The energy system in EEDx is heavily inspired by Hamon et al. (15), as shown in Figure 4(a). Formally, we note the energy level of the agent a at time t as $E_a(t)$. Energy can be gained by consuming food, which is done passively each time the agent's body overlaps with a cell containing food. If so, the agent receives an amount of energy corresponding to the energy content e_f of the food type f present in the cell, and the cell is emptied.



(a)



(b)

Figure 4: **(a)** Agents in EEDx have to maintain their energy levels in the positive range for a $\Delta_T^{reproduce}$ timesteps. If agents' energy stays negative for more that Δ_T^{die} timesteps they die. **(b)** An agent in EEDx is defined by (i) a sensory interface S defining how the local environment state is transcribed into neural inputs, (ii) a neural network model N defining how neural states are updated based on inputs provided by the sensory interface and (iii) a motor interface M defining how neural states are transcribed into motor actions. Red arrows illustrate energy costs from each module.

At each time step, agents lose energy corresponding to their operating costs. This amount is composed of a constant basal energy expenditure representing body maintenance costs and component-specific costs arising from the activity and structure of the different agent components, such as its neural network or locomotor system.

Agents can reproduce if they maintain their energy level positive for enough time and die if their energy level is in the negative range for too long. Formally, we note the time spent in the positive range at time t as $\Delta_T^+(t)$ (we omit the agent indexing for ease of notation here)and the time spent in the negative range as $\Delta_T^-(t)$. At each time step, $\Delta_T^+(t)$ ($\Delta_T^-(t)$) is incremented by one if the agent's energy level is positive (negative). When the the reproduction (death) time threshold is reached: $\Delta_T^+(t) > \Delta_T^{reproduction}(t)$ ($\Delta_T^-(t) > \Delta_T^{death}(t)$), the agent reproduce (die).

When reproducing, agent a makes a copy of their genotype $g_a \in G$ that is then mutated with mutation operator $\mathcal{M}$ and passed to its child: $g_c = \mathcal{M}(g_a)$. Genotypes encode two types of information: (i) the size of the agents represented by a real-valued positive number and (ii) parameters related to their neural model. The latter is left intentionally vague as it will depend on the model in use (see Neural Networks section). The mutation operator is left unspecified as it will depend on the agents neural model. In the case of continuous genotypes, i.e. $G \equiv \mathbb{R}^{dg}$, we provide a generalised mutation mechanism adding gaussian white noise $\epsilon \sim \mathcal{N}(0, \sigma)$ to randomly sampled locations of the genotype.

Importantly, by also allowing the size of the agent to vary through evolution, EEDx can be used to study scale effects on the development of sensory systems.

Sensory Interface

The sensory interface defines how the external local environment is transcribed into inputs to the agent's neural network. Formally, We note $S : S^S \times O \rightarrow I \times S^S \times \mathbb{R}_+$ the interface where S^S is the interface state, O is the observations state corresponding to the local available environmental states and I is the neural input space that will be sent to the neural interface. The last output corresponds to the energy costs of the sensory function.

EEDx provides a set of different sensory interfaces that can be adapted to different neural architectures, e.g. image-like inputs for convolutional neural networks. Here, we describe in more detail the interface used in the experiments in this paper which supports spatially embedded neural networks, i.e. networks whose neurons have locations in space.

We omit the agent index a for ease of notation and describe the interface for a single agent. We note $\mathbf{x} \in \mathbb{R}^2$ the absolute position of the agent and with $\overline{x} \in \Omega$ the cell coordinate associated with a continuous position $x \in \mathbb{R}^2$. Neurons' positions within the agent's body are given by $\{x_i \in [-1, 1]^2\}_{i<N}$ where N is the total number of neurons. Each neuron has an associated parameter $s_i \in \mathbb{R}^{n^C}$ defining its sensitivity to each of the environment chemicals. The input from the environment to neuron i is then given by the chemical concentrations in the cell the neuron lies multiplied by its sensitivity: $I_i(t) = \sum_c s_{i,c}^C \cdot C_c^t(\overline{\mathbf{x} + x_i})$. We also include internal inputs such that energy levels by treating it as a constant field.

Neural Networks

Neural networks in EEDx are defined by two main components: (i) an encoding, or developmental model, generating the parameters and initial state of the neural network from the agent's genotype and (ii) a neural model mapping the current state of the network and the neural inputs provided by the sensory interface to the updated neural state. Formally, let $D : G \rightarrow S^N$ denote the developmental model mapping genotypes $g \in G$ into network states $s^N \in S^N$ and $N : S^N \times I \rightarrow S^N$ denote the neural model updating the network state based on inputs $i \in I$ coming from the sensory interface. We use the term state to refer to both dynamical state, e.g neurons' activations, and often more static features, usually referred to as parameters, like connection weights. This formulation is very general and allows to model a wide range of neural networks. For instance, direct encodings (i.e. without developmental processes) can be modelled by having an encoding function D simply copying parameters from the genome into the network state. Also, since all the determinants of neural dynamics are embedded within the state, this formulation allows for networks with dynamic 'parameters', e.g. with plastic weights. The last output of the neural module N corresponds to its energy costs which can depend on its dynamic activity or its more static structure like its weights or the number of neurons.

Motor Interface

The motor interface defines how neural states are transcribed into motor actions defining how the agents will move. Formally, the motor interface $M : S^N \times S^M \to A \times S^M \times \mathbb{R}$ maps the agent neural state $s^N(t)$ onto the action space A corresponding to the agent's angular and linear velocity as a function of the current motor state $s^M \in S^M$. As for neural and sensory modules, the motor component also outputs an energy cost arising from its function.

Braitenberg Vehicles Braitenberg vehicles (6) are simple vehicles equipped with two wheels on each side of their body. Their forward kinematics is a function of each of the wheels speed creating differential drive. When coupled to spatially embedded neural networks, the speed of each wheel is set by summing all motor neurons activations within range of each wheel.

Ciliated Organisms One of the probable origins of neural circuits might have been for the control of ciliary locomotion (Mitchell; 17). Inspired by these works, we provide in EEDx a motor interface emulating simplified kinematics for a ciliated organism (illustrated in Figure 5). In this setting, agents' bodies are still represented as squares of side s, where s can vary between individuals. Neurons, here spatially embedded, have positions in space (in the $[-1, 1]$ square representing the agent's body). Here, motor neurons act as individual cilias producing thrust whose direction depends on the position of the neuron in the body. If a neuron is too far from the border of the agent then it cannot produce any thrust. Angular and linear velocity of the whole body are computed by aggregating all individual neurons thrusts depending on their position. One interesting effect of this embodiment is that, since neurons directly act as motors and that their position can change through mutations to the neural model, we have a case of brain-body coevolution. We also include a simplified interface without any rotation.

Developing Neural Networks

As stated in the introduction, developmental theories of early neural evolution hint at the sensorimotor origins of neurons in ciliated organisms, i.e. initially sensorimotor cells (both sensitive to external stimuli like light and equipped with motor apparatus, e.g. cilias (17)) differentiated through changes in the developmental regulatory programs into distinct subtypes: specialized sensory and motor cells connected through an axon. This division of labour event further allowed for the control of bigger motor apparatus (17) and the development of more complex sensory apparatus like eyes (1).

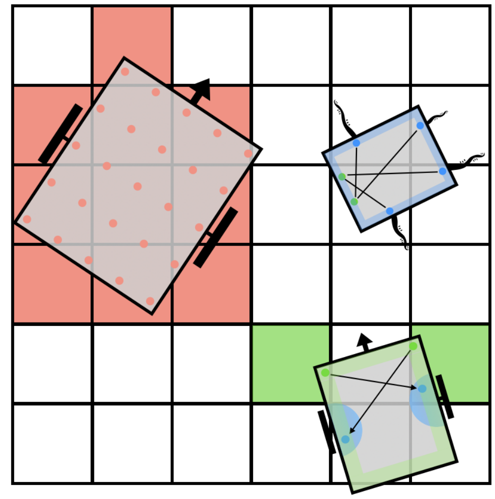


Figure 5: Agents in EEDx. On the left and bottom are Braitenberg vehicles like (6) agents. The left vehicle shows how bodies are approximated (which is independent of the motor or sensory interface): a fixed evenly spaced set of points is computed from the agent position and heading, red cells show the set of cells covered by the agent where at least one of the body points lie into. The right vehicle illustrates how the spatially embedded sensory model interface neurons and the environment. Sensory neurons (green dots) can sense chemicals concentrations in the cells they are (see green cells) at the condition they lie on the border of the agent (green shaded area in the agent). Motor neurons (blue dots) activate motors if in their respective activation areas (blue semi-circles). The top-right agent illustrates the ciliated motor interface. Motor neurons (blue), if on the border of the organism (blue region), emulate a cilia like component whose beating rate, and so the force it applies, is a function of the neuron's activation.

Inspired by the central role of developmental regulatory dynamics in these phylogenetic refinement dynamics, we propose RAND (Regulation based Artificial NeuroDevelopment), a new model for the development of neural networks based on abstract regulatory dynamics directing the migration, wiring and dynamics of artificial neurons. Like NDPs (24), RAND models the development of neural networks in a self-assembling manner where neurons communicate and organize in a time extended process. RAND differs in that neurons are spatially embedded, i.e. have positions in space,

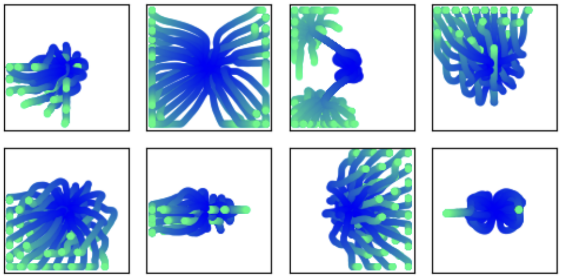


Figure 6: Migration patterns obtained from randomly sampled RAND parameters.

a desirable properties to model local communication, realistic energetic costs and body-brain co-evolution.

In our model, neurons have an internal state which can be paralleled to the gene expression profile of biological cells. At the onset of development, our artificial neurons are undifferentiated, meaning they all have the same state, and they start out with random positions close to the center of the body where the body is the $[-1, 1]^2$ square. Development follows two phases: (i) migration, where neurons migrate differentially following different molecular signals according to their internal state and (ii) wiring, where neurons connect to each other forming the final network.

Regulation and Communication Let $s_i(t) \in \mathbb{R}^{ds}$ be the state of neuron i at time t and $x_i \in \mathbb{R}^2$ its position. The internal state of neurons is influenced by (i) intrinsic regulation dynamics encoded by the state regulation matrix $W_{in} \in \mathbb{R}^{ds \times ds}$ and (ii) extrinsic influence from their local environment encoded by the matrix $W_{ex} \in \mathbb{R}^{sg \times m}$ where m is the number of different molecules composing the external milieu. The external milieu is defined by (i) a fixed morphogen field $\tilde{M} : \mathbb{R}^2 \to \mathbb{R}^m$ and (ii) signalling molecules emitted by neurons giving the complete field:

$$M(x) = \left[\tilde{M}(x) \Big|\Big| \sum_i \tilde{m}_i \cdot e^{-\frac{||x_i - x||}{\sigma^2}} \right] \tag{7}$$

With $[\cdot||\cdot]$ the concatenation operator. $\tilde{m}_i$ is decoded from a subset of neurons' internal states via a linear projection. Neurons' internal states follow the dynamical system:

$$\tau \frac{ds_i}{dt} = -s_i + \sigma(W_{in} s_i + W_{ex} M(x_i)) \tag{8}$$

Where $\tau \in \mathbb{R}^{ds}$ is a gene specific time constant and σ is a transfer function (e.g. the hyperbolic tangent).

Migration Given M is differentiable, migration dynamics are given by the following differential equation:

$$\frac{dx_i}{dt} = -\nabla \langle \lambda_i, \tilde{M}(x_i) \rangle \tag{9}$$

Where $\lambda_i \in \mathbb{R}^m$ is directly encoded in neurons' internal states and encode attraction/repulsion to the different molecular concentrations given by the field $\tilde{M}$, i.e. how strongly and in which direction it should follow the different fields' gradients. By following the perturbed field $\tilde{M}$, attraction/repulsion mechanisms between neurons can be evolved. We enforce boundary conditions by clipping positions into the $[-1, 1]$ body domain. Figure 6 shows migration patterns from randomly sampled parameters.

Wiring After neurons have reached their position at the end of a fixed developmental time T_{dev}, the synaptogenesis phase can take place. We model this process using the Genetic Connectome Model (GCM), which models how the interaction between different synapse proteins determine the synaptogenesis process (2; 18). Furthermore, this model has been shown to promote structured connectomes (4) and has already been used in artificial evolution settings (3). The GCM determines the final connectome $W \in \mathbb{R}^{\mathbb{N} \times \mathbb{N}}$ as a function of the neurons' synaptic proteins expression given by the matrix $X \in \mathbb{R}^{N \times s}$ where s is the number of different synapse proteins and a set of wiring rules $O \equiv \{O_r \in \mathbb{R}^{\sim \times \sim}\}$ defining interactions between proteins:

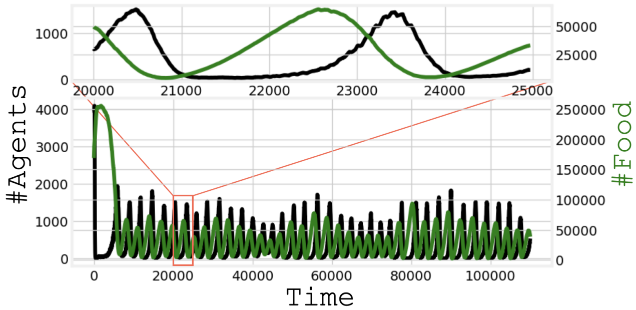
$$W = \sum_r X O_r X^T \tag{10}$$

In our model, synapse proteins of neuron i, X_i, is directly encoded in part of its internal state s_i. Overall, the complete set of evolvable parameters is $\{W_{in}, W_{ex}, \tau, O\}$.

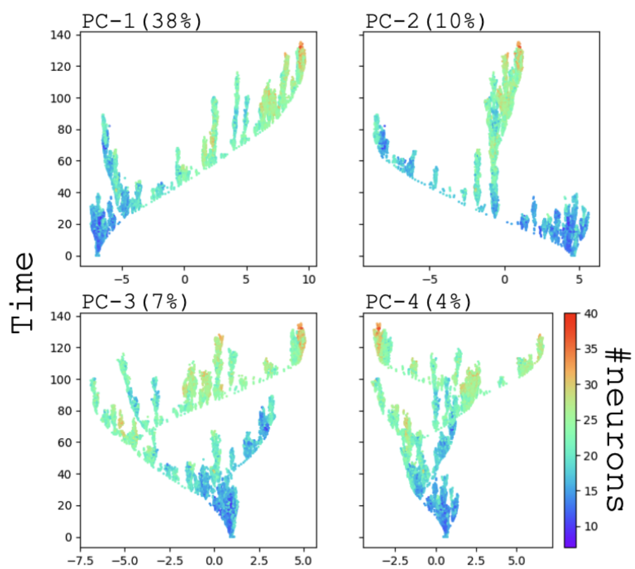
Figure 7: We compare times to simulate one million timesteps in different conditions and with different hardware setups. The three plots to the left show simulation time scaling with the number of agents for different environment sizes (grids of size $S \times S$) and different device counts (NVIDIA RTX 6000 Ada generation GPUs). The right plot shows performance scaling with agents' neural networks sizes for different devices counts.

Experiments

System Profiling We tested the scaling behaviour of the system in terms of computation time in different settings. Results are displayed in Figure 7. Being implemented in JAX (5), EEDx can take advantage of hardware acceleration. Moreover, EEDx provides automatic support for multi-GPU setups by distributing the population of agents across



(a)



(b)

Figure 8: EEDx simulation data. **(a)** Predator-prey like oscillations are observed in a large proportion of simulations. **(b)** Evolutionary trees produced from simulation data following the methodology presented in Plantec et al. (26). The x-axis represent one of the four first principal components of extracted from the set of all genotypes that appeared during the simulation. Color codes for the number if neurons in the associated phenotypic neural network.

available devices. While it introduces overheads because of communication between devices resulting in slower simulation with small environments being simulated, it shows better scaling for large environments.

Simulations We validate our system by analysing simple simulations in which agents' neural networks were encoded with RAND. The environment is composed of a single food species growing strictly locally ($D^{min} = D^{max} = 1$) and emitting a single chemical species. In a large proportion of runs we observe population oscillations coupled to resource levels oscillations reminiscent of Lotka-Volterra equations (20; 37) describing population dynamics in two-species predator-prey ecosystems (see Figure 8(a)). Using the methodology presented in (26), we generate "evolutionary trees" showing the genotype trajectories through time (Figure 8(b)). Interesting patterns can be observed, especially branching patterns hinting at speciation-like events. We can also indicate on these trees properties of the developed phenotypic neural networks, here the number of neurons. We can see an increasing trend through evolution, which finds a natural explanation in this simulation: all motor neurons can only produce a maximum amount of force; when this amount is set to be low, more motor neurons are needed to produce more thrust allowing more efficient foraging, thus creating a pressure for larger networks.

Discussion and Conclusion

The emergence of cognition in evolution refers to the development of information processing and agency, both allowed by the rise of nervous systems, sensory organs, and motor control. It required new developmental pathways, specialised neural and sensory cells, and internal coordination to integrate stimuli and produce coherent responses. Cognition evolved under ecological pressures, such as predation, which favour flexible and anticipatory behaviour. During the Cambrian explosion, many animal lineages rapidly developed complex body plans, centralised nervous systems, and sensory structures, making this period a likely threshold for the first major emergence of cognitive traits.

This event marks a key evolutionary transition in which internal processing and environmental responsiveness became tightly coupled in behaviourally sophisticated organisms. However, how the different requirements allowed this transition to occur is far from obvious. Due to the tangled eco-evo-devo nature of the problem, theoretical models that explicitly consider the construction of neural networks within bodies and their interaction with spatial environments might be the only way to understand the emergence of cognitive agents fully. This is the main motivation for the work presented here. In this context, EEDx offers a novel software platform to achieve these goals.

By integrating expressive, code-free ecological modelling with developmental neural networks, EEDx provides a flexible environment to simulate how cognitive traits may emerge from sensorimotor foundations under varying selective pressures. Our approach bridges theoretical frameworks by allowing controlled experimentation at the intersection of ontogeny and ecological interaction, offering new insights into the conditions that may have driven major transitions in neural complexity.

Acknowledgements

Funded by the European Union (ERC, GROW-AI, 101045094). Views and opinions expressed are however those of the authors only and do not necessarily reflect those of the European Union or the European Research Council. Neither the European Union nor the granting authority can be held responsible for them. RS acknowledges the support of a grant AGAUR 2021 SGR0075 and by the Santa Fe Institute.

References

Arendt, D., Hausen, H., and Purschke, G. (2009). The 'division of labour' model of eye evolution. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 364(1531):2809–2817. Publisher: Royal Society.

Barabási, D. L. and Barabási, A.-L. (2020). A Genetic Model of the Connectome. *Neuron*, 105(3):435–445.e5.

Barabási, D. L., Beynon, T., Katona, , and Perez-Nieves, N. (2023). Complex computation from developmental priors. *Nature Communications*, 14(1):2226. Publisher: Nature Publishing Group.

Barabási, D. L. and Czégel, D. (2021). Constructing graphs from genetic encodings. *Scientific Reports*, 11(1):13270. Publisher: Nature Publishing Group.

Bradbury, J., Frostig, R., Hawkins, P., Johnson, M. J., Leary, C., Maclaurin, D., Necula, G., Paszke, A., VanderPlas, J., Wanderman-Milne, S., and Zhang, Q. (2018). JAX: composable transformations of Python+NumPy programs.

Braitenberg, V. (1986). *Vehicles: Experiments in Synthetic Psychology*. MIT Press. Google-Books-ID: 7KkUAT_q_sQC.

Briggs, D. E. (2015). The cambrian explosion. *Current Biology*, 25(19):R864–R868.

Budd, G. E. and Jensen, S. (2017). The origin of the animals and a 'Savannah' hypothesis for early bilaterian evolution. *Biological Reviews*, 92(1):446–473. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/brv.12239.

Chaturvedi, S., El-Gazzar, A., and Gerven, M. v. (2024). Foragax: An Agent-Based Modelling Framework Based on JAX. arXiv:2409.06345 [cs].

de Vladar, H. P. and Szathmáry, E. (2015). Neuronal boost to evolutionary dynamics. *Interface focus*, 5(6):20150074.

Erwin, D. H. (2020). The origin of animal body plans: a view from fossil evidence and the regulatory genome. *Development*, 147(4):dev182899.

Erwin, D. H., Laflamme, M., Tweedt, S. M., Sperling, E. A., Pisani, D., and Peterson, K. J. (2011). The cambrian conundrum: early divergence and later ecological success in the early history of animals. *science*, 334(6059):1091–1097.

Fedonkin, M. A. (2007). *The rise of animals: evolution and diversification of the kingdom Animalia*. JHU Press.

Ginsburg, S. and Jablonka, E. (2021). Evolutionary transitions in learning and cognition. *Philosophical Transactions of the Royal Society B*, 376(1821):20190766.

Hamon, G., Nisioti, E., and Moulin-Frier, C. (2023). Eco-evolutionary Dynamics of Non-episodic Neuroevolution in Large Multi-agent Environments. arXiv:2302.09334 [cs].

Jablonka, E. and Lamb, M. J. (2006). The evolution of information in the major transitions. *Journal of theoretical biology*, 239(2):236–246.

Jékely, G. (2010). Origin and early evolution of neural circuits for the control of ciliary locomotion. *Proceedings of the Royal Society B: Biological Sciences*, 278(1707):914–922. Publisher: Royal Society.

Kovács, I. A., Barabási, D. L., and Barabási, A.-L. (2020). Uncovering the genetic blueprint of the C. elegans nervous system. *Proceedings of the National Academy of Sciences*, 117(52):33570–33577. Publisher: Proceedings of the National Academy of Sciences.

Llinás, R. R. (2002). *I of the vortex: From neurons to self*. MIT press.

Lotka, A. J. (1927). Fluctuations in the Abundance of a Species considered Mathematically. *Nature*, 119(2983):12–12. Publisher: Nature Publishing Group.

Mitchell, D. R. The evolution of eukaryotic cilia and flagella as motile and sensory organelles. *Eukaryotic membranes and cytoskeleton: Origins and evolution*, pages 130–140.

Monk, T. and Paulin, M. G. (2014). Predation and the Origin of Neurones. *Brain, Behavior and Evolution*, 84(4):246–261.

Najarro, E., Sudhakaran, S., Glanois, C., and Risi, S. (2022). HyperNCA: Growing Developmental Networks with Neural Cellular Automata. arXiv:2204.11674 [cs].

Najarro, E., Sudhakaran, S., and Risi, S. (2023). Towards Self-Assembling Artificial Neural Networks through Neural Developmental Programs. MIT Press.

Paulin, M. G. and Cahill-Lane, J. (2021). Events in Early Nervous System Evolution. *Topics in Cognitive Science*, 13(1):25–44. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/tops.12461.

Plantec, E., Hamon, G., Etcheverry, M., Chan, B. W.-C., Oudeyer, P.-Y., and Moulin-Frier, C. (2025). Flow-Lenia: Emergent Evolutionary Dynamics in Mass Conservative Continuous Cellular Automata. *Artificial Life*, pages 1–21.

Plantec, E., Pedersen, J. W., Montero, M. L., Nisioti, E., and Risi, S. (2024). Evolving Self-Assembling Neural Networks: From Spontaneous Activity to Experience-Dependent Learning. MIT Press.

Plotnick, R. E., Dornbos, S. Q., and Chen, J. (2010). Information landscapes and sensory ecology of the Cambrian Radiation. *Paleobiology*, 36(2):303–317.

Podlozhnyuk, V. (2007). FFT-based 2D convolution.

Smith, M. P. and Harper, D. A. T. (2013). Causes of the Cambrian Explosion. *Science*, 341(6152):1355–1356. Publisher: American Association for the Advancement of Science.

Solé, R., Fernandez, P., and Kauffman, S. A. (2003). Adaptive walks in a gene network model of morphogenesis: insights into the cambrian explosion. *International journal of developmental biology*, 47(1821):685–693.

Solé, R., Moses, M., and Forrest, S. (2019). Liquid brains, solid brains. *Philosophical Transactions of the Royal Society B*, 374(1774):20190040.

Stanley, K., D'Ambrosio, D., and Gauci, J. (2009). A Hypercube-Based Encoding for Evolving Large-Scale Neural Networks. *Artificial Life*, 15(2):185–212.

Szathmáry, E. (2015). Toward major evolutionary transitions theory 2.0. *Proceedings of the National Academy of Sciences*, 112(33):10104–10111.

Tiwary, K., Young, A., Tasneem, Z., Klinghoffer, T., Dave, A., Poggio, T., Nilsson, D.-E., Cheung, B., and Raskar, R. (2025). What if Eye...? Computationally Recreating Vision Evolution. arXiv:2501.15001 [cs].

Vickers, N. J. (2000). Mechanisms of animal navigation in odor plumes. *The Biological Bulletin*, 198(2):203–212.

Volterra, V. (1926). Fluctuations in the Abundance of a Species considered Mathematically1. *Nature*, 118(2972):558–560. Publisher: Nature Publishing Group.